

Notes: Caliendo and Rossi-Hansberg (QJE, 2012)

The Impact of Trade on Organization and Productivity

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1 Big picture

- **Question.** How does international trade change the internal organization of firms, and how do those organizational changes feed back into productivity and gains from trade?
- **Core idea.** A firm's productivity is not purely exogenous. It depends on how the firm organizes production: how many management layers it has, how much knowledge workers and managers acquire, and how wide each manager's span of control is.
- **Mechanism.** Trade changes firm scale. Exporters expand output and may add management layers. Nonexporters shrink and may drop layers. These discrete organizational changes generate heterogeneous productivity responses.
- **Main theoretical contribution.** The paper embeds a Garicano-style knowledge hierarchy inside a Melitz-style monopolistic competition trade environment, replacing exogenous productivity heterogeneity with productivity generated by endogenous organization.
- **Main quantitative result.** In the calibrated U.S. manufacturing economy, moving from autarky to 2002 openness raises productivity of the marginal exporter by about 1% and revenue productivity by about 1.8%. Endogenous organization raises gains from trade by about 41% relative to the comparable Melitz-style benchmark.
- **Why this matters.** Standard heterogeneous-firm trade models usually treat firms as black-box production technologies. This paper opens the black box and shows that market access affects the firm's hierarchy, wage distribution, knowledge allocation, and productivity.

2 Incremental contribution of the paper

- **Endogenous productivity from organization.** Instead of drawing productivity directly from an exogenous distribution, the model draws firm demand and lets productivity arise from the cost-minimizing organization selected by the firm.
- **Trade and hierarchy.** The paper gives a precise theory of how exporting changes the number of layers, the knowledge of employees in each layer, the span of control, and managerial autonomy.
- **Microfoundation of average cost curves.** The paper derives nonmonotone, lower-envelope average cost curves from a fully specified hierarchy model. These curves resemble classic average cost curves, but every point corresponds to an explicit organization.
- **Melitz as a limiting case.** When knowledge is costless or unnecessary, organization becomes irrelevant and the model collapses to a standard constant-marginal-cost trade model with exogenous productivity-like differences.
- **Heterogeneous productivity effects of trade.** Exporting does not mechanically raise every firm's physical productivity. Some firms gain, some lose, depending on where they are relative to the minimum efficient scale of their current layer structure.
- **Quantitative gains from organization.** By calibrating to U.S. manufacturing, the paper shows that endogenous organizational adjustment makes the welfare gains from trade materially larger than in a standard model.

- **Testable organizational predictions.** Exporters should add layers; nonexporters should drop layers. Exporters that add layers should employ more people in each layer but require less knowledge per employee in comparable layers. Nonexporters that drop layers should show the reverse.

3 Data, calibration, and empirical targets

3.1 Calibration target and scope

The quantitative exercise is a two-country symmetric model calibrated to the U.S. manufacturing sector around 2002. The model bundles manufacturing with the part of the education sector needed to train workers because knowledge acquisition is a real input in production.

The paper calibrates the model to match moments that summarize the size and openness of U.S. manufacturing and the knowledge-training component of the economy. The key data targets are:

- share of manufacturing firms that export, about 18%;
- average firm size, about 45 workers;
- total expenditure, normalized to match the manufacturing bundle;
- share of expenditure on domestic goods, about 78.9%;
- share of education employees, about 11.8%;
- Pareto coefficient of the firm size distribution.

3.2 Parameterization

The model uses standard values from the trade literature for some parameters: elasticity of substitution $\sigma = 3.8$, iceberg trade cost $\tau = 1.3$, and exit rate $\delta = 10\%$. The domestic fixed production cost is normalized at $f_{ii} = 1.1$. The remaining parameters are chosen to match the target moments.

Important calibrated parameters include:

- $A = 0.26$: baseline output per production possibility;
- $f^E = 35.1$: entry cost;
- $f_{ij} = 5.4$: fixed export cost;
- $\gamma = 0.9$: Pareto demand-shock parameter;
- $c/\lambda = 0.225$: cost of acquiring knowledge relative to the problem distribution;
- $h = 0.26$: communication cost, or time cost of transmitting unsolved problems upward.

Interpretation. The calibration is not an empirical test in the reduced-form sense. It is a structural quantitative exercise: choose parameters so the model matches basic U.S. manufacturing moments, then use the model to study counterfactual trade liberalization and organizational adjustment.

4 Model environment

4.1 Preferences and product demand

Consumers have CES preferences over differentiated varieties. Each potential product has a demand shifter α , drawn after the entrepreneur pays an entry/design cost. The demand shifter is the only exogenous source of firm heterogeneity in the closed-economy baseline.

The key point is that heterogeneity in demand becomes heterogeneity in organization and productivity. High-demand firms choose larger scale, more complex hierarchies, and different knowledge allocations.

4.2 Production as problem solving

Production requires both labor time and knowledge. Workers generate production possibilities, but output is realized only if the associated production problem is solved.

Problems are drawn from a distribution $F(z) = 1 - e^{-\lambda z}$. Problems near zero are common; higher z problems are rarer. Acquiring knowledge is costly: learning an interval of problems of length z costs wcz . Communication is also costly: when a worker cannot solve a problem, sending it to a manager consumes h units of managerial time.

4.3 Hierarchy and layers

A firm with highest layer L has $L + 1$ layers: workers at layer 0 and managers at layers $1, \dots, L$. The top manager is the entrepreneur. Workers learn routine problems. Higher-level managers learn increasingly rare exceptions.

Let z_L^ℓ denote the knowledge of agents in layer ℓ of a firm with highest layer L . Let n_L^ℓ denote the number of agents in that layer. The top layer has one agent:

$$n_L^L = 1. \tag{1}$$

The span of control is shaped by lower-layer knowledge. If lower layers solve more problems, fewer questions move upward, so each manager can supervise more agents.

4.4 Cost minimization

The firm chooses the number of layers, employment in each layer, and knowledge in each layer to minimize the cost of producing quantity q :

$$C(q, w) = \min_{L \geq 0} C_L(q, w), \tag{2}$$

where $C_L(q, w)$ is the minimum variable cost of producing q with $L + 1$ layers.

For a given layer structure, the firm solves:

$$C_L(q, w) = \min_{\{n_L^\ell, z_L^\ell\}_{\ell=0}^L} \sum_{\ell=0}^L n_L^\ell w (cz_L^\ell + 1), \tag{3}$$

subject to producing at least q , satisfying communication constraints, and assigning one full unit of time to the entrepreneur.

Interpretation. Adding a layer is like paying a larger fixed organizational cost to reduce marginal cost at larger scale. The firm adds management layers only when output is large enough to justify the extra management structure.

5 Cost-function logic and propositions

5.1 Proposition 1: within a fixed number of layers

Given a fixed number of layers L , as output q rises:

- marginal cost rises;
- knowledge rises in all layers;
- the number of employees rises in all non-top layers;
- managers' span of control rises;
- the entrepreneur's autonomy falls.

The average cost curve for a fixed layer count is convex. It falls at first because fixed organizational capacity is spread over more output, reaches a minimum efficient scale (MES), and then rises because scaling a fixed hierarchy becomes increasingly costly.

5.2 Proposition 2: across layer choices

As the firm switches from L to $L + 1$ layers:

- the minimum average cost associated with the new layer count is lower;
- the output level at which average cost is minimized is higher;
- marginal cost falls discontinuously;
- knowledge and span of control in comparable layers fall discontinuously;
- employment in each layer rises discontinuously.

Economic message. Adding a management layer lets the firm economize on knowledge in lower layers by leveraging higher-level problem solvers. The firm becomes larger, more specialized, and less centralized.

5.3 Productivity measure

The main productivity measure is the inverse of variable average cost with wages normalized to one:

$$a(q) = \frac{q}{C(q, 1)} = \frac{1}{AC(q, 1)}. \quad (4)$$

This measure treats knowledge acquisition and labor time as real inputs. It abstracts from fixed entry and fixed export costs when measuring variable productivity.

5.4 Proposition 3: the Melitz limit

When $c/\lambda \rightarrow 0$, knowledge is either unnecessary or costless. Organization no longer has an economic purpose. If self-employment is ruled out, the cost function becomes:

$$C(q, w) = \frac{qw}{A}, \quad AC(q, w) = MC(q, w) = \frac{w}{A}, \quad (5)$$

and productivity becomes simply A . This is isomorphic to the standard heterogeneous-firm trade model with constant marginal cost.

Interpretation. The standard model is nested inside the richer organizational model. The difference is that, when knowledge is costly, productivity varies with scale and organization.

6 Profit maximization and equilibrium

6.1 Firm pricing

Given demand draw α , the firm chooses output to maximize operating profit:

$$\pi(\alpha) = \max_{q \geq 0} \{p(\alpha)q - C(q, w) - wf\}. \quad (6)$$

With CES demand, the optimal price is a constant markup over marginal cost:

$$p(\alpha) = \frac{\sigma}{\sigma - 1} MC(q(\alpha), w). \quad (7)$$

Unlike in the standard model, marginal cost itself depends on output because output determines the firm's optimal organization.

6.2 Proposition 4: quantity, revenue, and price

For a given number of layers, $q(\alpha)$ and revenue rise continuously with α . When the firm adds a layer, quantity and revenue jump upward. Price rises with α within a layer but falls discontinuously when a new layer is added because the added layer lowers marginal cost.

Economic message. Larger firms do not necessarily charge higher or lower prices. The price schedule has discontinuities because firms discretely reorganize.

6.3 Proposition 5: profit concavity

For a given number of layers, the profit function is strictly concave in output. The firm can therefore solve local profit maximization by layer, and then choose the best layer count globally.

6.4 Closed-economy equilibrium

Free entry and zero-profit cutoff determine which demand draws operate. The marginal active firm earns zero profit. The equilibrium includes:

- cutoff demand $\bar{\alpha}$;
- wage w ;

- total revenue R ;
- mass of operating firms M .

Proposition 6 establishes a unique stationary equilibrium.

6.5 Proposition 7: population size

A larger population raises wages, prices, revenue, and profits, but it does not change the entry threshold, quantities produced, or organizational structure. This result clarifies that scale effects from market size alone are not enough; trade costs and export fixed costs matter because they change the set of firms and the scale of individual exporters.

7 Open economy and trade

7.1 Trade costs and export choice

The open economy has two countries. Firms can sell domestically and, after paying a fixed export cost f_{ij} , abroad. Iceberg trade costs $\tau_{ij} > 1$ imply that delivering one unit abroad requires producing τ_{ij} units.

A key difference from Melitz-style models is that domestic and export sales are not separable in the cost function. Total quantity determines marginal cost and organization:

$$q_i(\alpha) = q_{ii}(\alpha) + q_{ij}(\alpha). \quad (8)$$

Thus, exporting can affect the cost of producing domestic sales by causing a firm to reorganize.

7.2 Proposition 8: trade liberalization

A bilateral trade liberalization increases welfare in both countries. It causes:

- all exporters to produce more;
- all nonexporters to produce less.

The effect on productivity is heterogeneous. Exporters tend to gain productivity on average, especially when export scale pushes them into a more efficient hierarchy. Nonexporters may gain or lose depending on how their reduced scale relates to their MES.

7.3 Corollary 9: trade and layers

Trade liberalization weakly increases the number of management layers of exporters and weakly decreases the number of layers of nonexporters.

Economic message. Exporting expands the firm and makes a more elaborate hierarchy worthwhile. Competition and selection shrink nonexporters and can make flatter organization optimal.

8 Quantitative findings

8.1 Productivity by demand draw

The calibrated model shows that exporters are more productive than nonexporters on average. The marginal exporter experiences a discrete productivity increase because exporting pushes the firm across a reorganization threshold. However, productivity responses are not uniform: some exporters and some nonexporters can lose physical productivity depending on their position relative to MES.

8.2 Gains from trade

In the calibrated organizational model, moving from autarky to the 2002 level of openness yields an 8.1% welfare gain. The corresponding Melitz-limit economy yields a 5.7% gain. Thus, organizational adjustment raises gains from trade by about 2.4 percentage points, or 41%.

Interpretation. Part of the gains from trade comes from within-firm restructuring, not just selection across firms or access to more varieties.

8.3 Distributional implications inside firms

Opening to trade changes the distribution of knowledge and income:

- exporters add layers, employ more workers and managers, and often reduce knowledge per worker in comparable lower layers;
- nonexporters shrink and may remove layers, increasing the knowledge needed by remaining workers in comparable layers;
- average income rises because wages rise in the open economy;
- the open economy has more mass at high productivity levels.

8.4 Autonomy and decentralization

The model defines the entrepreneur's autonomy as the share of total firm knowledge held by the entrepreneur:

$$\frac{z_L^L}{Z_L^L}. \quad (9)$$

Higher demand firms are less centralized. Trade makes nonexporters more centralized and exporters less centralized.

8.5 Export fixed costs

When fixed export costs are zero, all firms export. In a symmetric world, this is similar to a market-size increase and leaves firm quantities and organization unchanged. When export fixed costs are infinite, no firm exports and the economy is autarkic. Productivity gains are therefore zero at both extremes and positive for intermediate fixed export costs.

8.6 Knowledge and communication technology

The model's organizational channel depends on the cost of knowledge and the cost of communication.

- Higher knowledge cost c makes organizational economizing more valuable. Trade-induced expansion by exporters then produces larger organizational and productivity effects.
- Higher communication cost h reduces managerial span of control, makes new layers less attractive, and weakens the organizational gains from trade.

9 Appendices and proof logic

9.1 Appendix I: alternative productivity measures

The main text uses physical productivity $q/C(q, 1)$. Appendix I studies productivity measures closer to common empirical variables:

- revenue productivity, $r(\alpha)/C(q(\alpha), 1)$;
- labor productivity, $q(\alpha)/n(\alpha)$;
- revenue labor productivity, $r(\alpha)/n(\alpha)$.

The central message is that exporters are more productive than nonexporters on average under all measures, but revenue-based measures can overstate productivity gains because they include price increases caused by higher wages and markups. The marginal exporter gains about 1% in physical productivity, about 1.82% in revenue productivity, about 2% in labor productivity, and about 2.83% in revenue labor productivity.

9.2 Appendix II: proof structure

The appendix proves the formal propositions. The main proof device is the characterization of how knowledge constraints bind and how the Lagrange multiplier on the output constraint moves with output. Assumption 1 guarantees that knowledge remains positive in relevant layers. Figure B1 illustrates the equilibrium cutoff and free-entry schedules used in proving existence and uniqueness.

10 Tables and figures

10.1 Cost curves and organizational technology

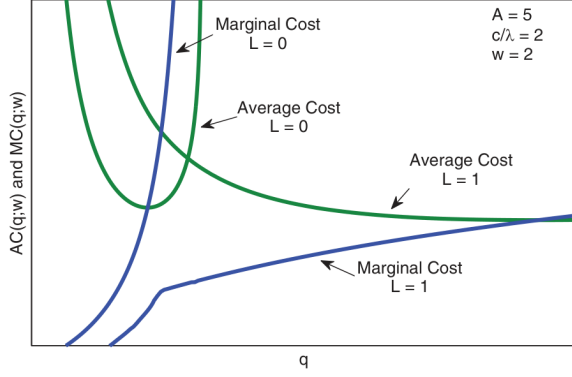


FIGURE I
Average and Marginal Cost, $L = \{0, 1\}$

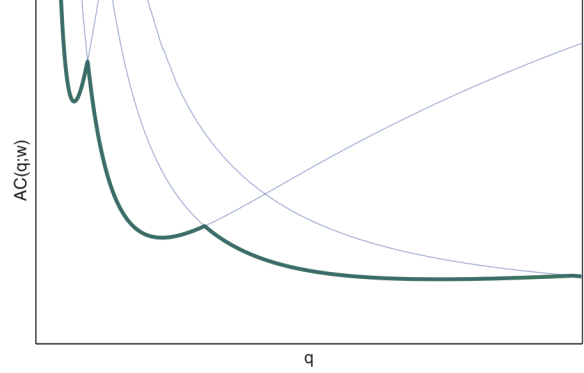


FIGURE II
The Cost Function

Figure I. Average and marginal cost with one versus two layers.

Figure II. The lower-envelope cost function across layer choices.

Read. Figure I shows that with a fixed number of layers average cost is convex and marginal cost rises with output. Figure II shows the lower envelope across layer-specific average cost curves; the envelope is not globally convex or monotone because firms discretely switch layers.

Economic message. Organization creates scale-dependent productivity. Adding a layer is costly, but it can lower marginal and average cost at higher scale by economizing on knowledge in lower layers.

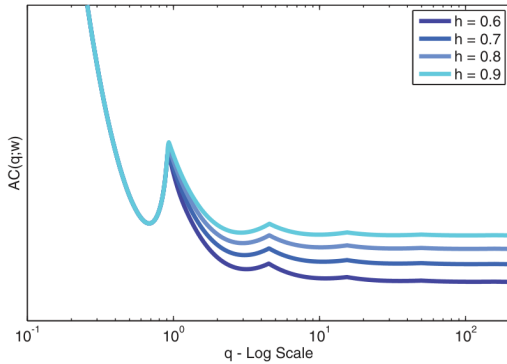


FIGURE III
Changing h ($\xi = 1$)

Figure III. Average cost as communication cost h changes.

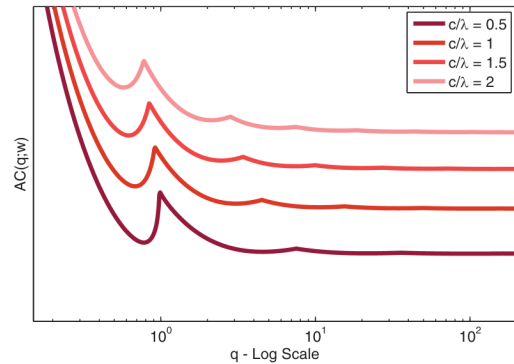


FIGURE IV
Changing ξ ($h = 0.7$)

Figure IV. Average cost as knowledge cost c/λ changes.

Read. Higher communication cost raises average cost and flattens the benefit of adding layers. Higher knowledge cost also raises average costs, especially for organized firms that rely heavily on trained agents.

Economic message. The gains from hierarchy are technological. Better communication and cheaper knowledge acquisition make organizational complexity more useful.

10.2 Calibration and productivity response

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TABLE I
CALIBRATED MOMENTS

Moments	Data	Model
Share of firms that export	18.0	17.53
Average size of firms	45.2	45.44
Total expenditure	5.1	5.10
Share of expenditure on domestic goods	78.9	74.94
Share of education employees	11.8	11.85
Size distribution of firms, pareto coefficient	-1.095	-1.094

Table I. Calibration targets and model fit.

TABLE II
PARAMETER VALUES

Parameters	A	f^E	f_{ij}	γ	$\frac{\varepsilon}{\lambda}$	h
Values	0.26	35.1	5.4	0.9	0.225	0.26

Table II. Calibrated parameter values.

Read. Table I shows that the model closely matches the export share, average firm size, total expenditure, education-employment share, and Pareto coefficient. Table II reports the calibrated technology, entry, export, knowledge, and communication parameters.

Economic message. The calibration anchors the model in U.S. manufacturing and makes the organizational channel quantitatively interpretable rather than purely theoretical.

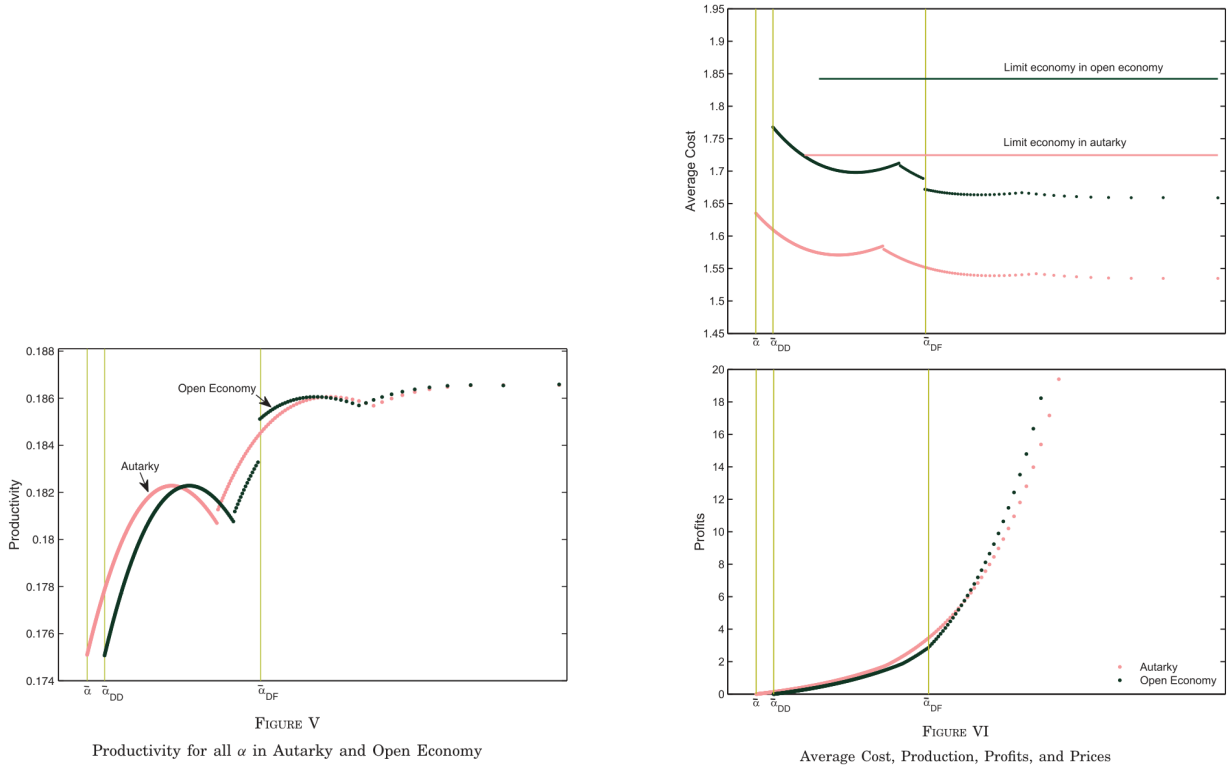


Figure V. Productivity across demand draws in autarky and openness.

Figure VI, part 1. Average costs and profits in autarky and openness.

Read. Figure V shows productivity as a nonmonotone function of demand because firms cross organizational thresholds. Figure VI part 1 shows higher average costs in the open economy due to higher wages, but higher profits for large exporters.

Economic message. Exporting changes firm productivity through reorganization. The productivity jump at the export threshold is a key source of within-firm gains.

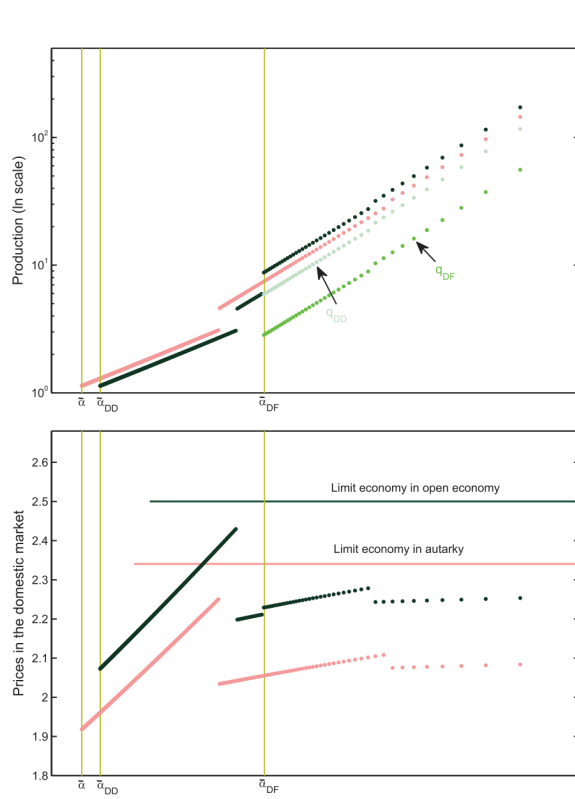


FIGURE VI
Continued

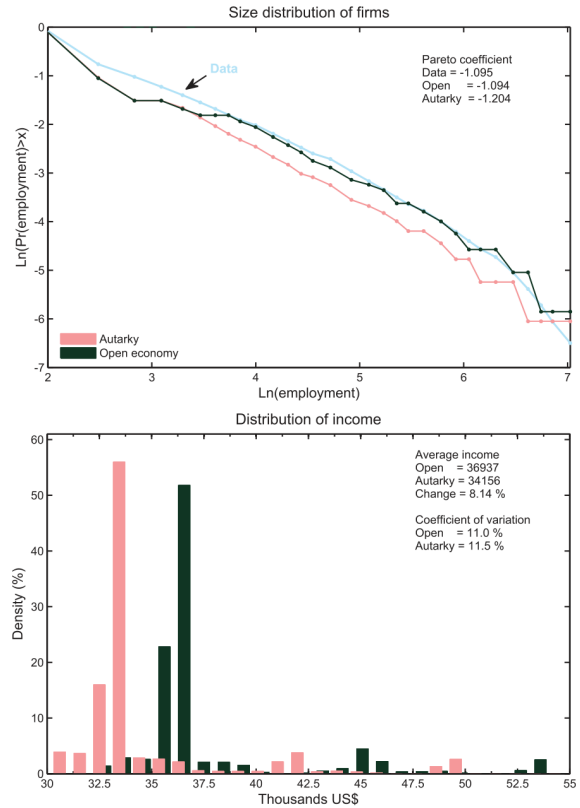


FIGURE VII
Firm Size, Knowledge, Income, and Productivity Distributions

Figure VI, part 2. Production and domestic prices.

Figure VII, part 1. Firm size and income distributions.

Read. Figure VI part 2 shows that exporters produce more overall, while domestic production can fall relative to autarky. Prices move with marginal cost and layer switches. Figure VII part 1 shows that openness thickens the right tail of the firm-size distribution and shifts the income distribution to the right.

Economic message. Trade reallocates scale toward exporters and changes both the extensive distribution of firm sizes and the intensive organization of each firm.

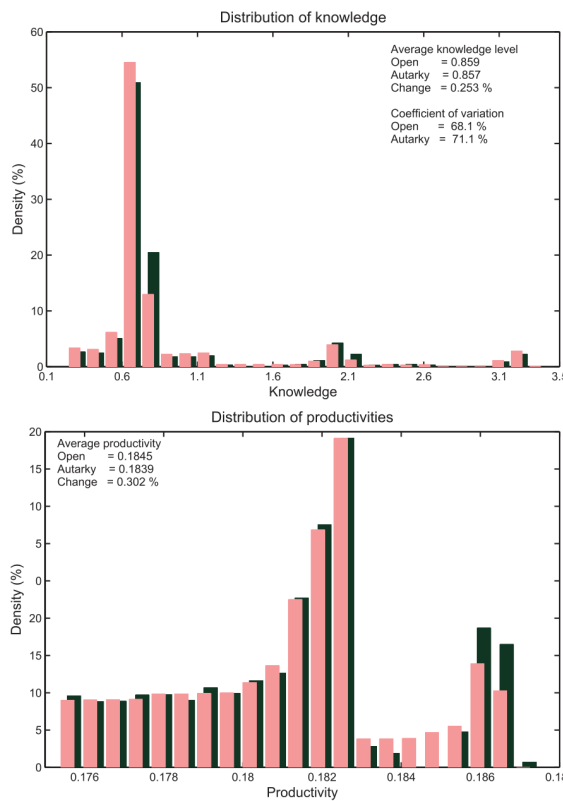


FIGURE VII
Continued

Figure VII, part 2. Knowledge and productivity distributions.

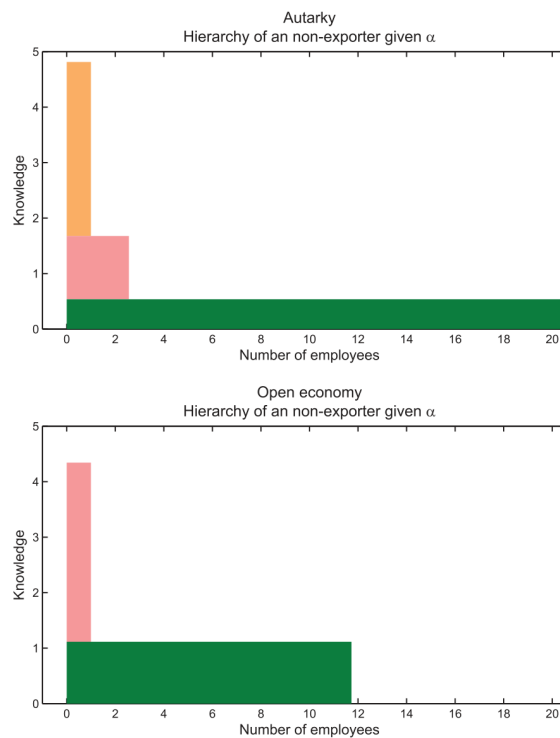


FIGURE VIII
Changes in the Organizational Design after a Bilateral Trade Liberalization

Figure VIII, part 1. Nonexporter hierarchy before and after trade.

Read. Figure VII part 2 shows more workers with low knowledge but also more intermediate managers in the open economy, reflecting added layers among exporters. Figure VIII part 1 shows a nonexporter dropping a layer and shrinking employment after trade.

Economic message. Trade changes within-firm knowledge allocation. Nonexporters become smaller and more centralized when they lose scale.

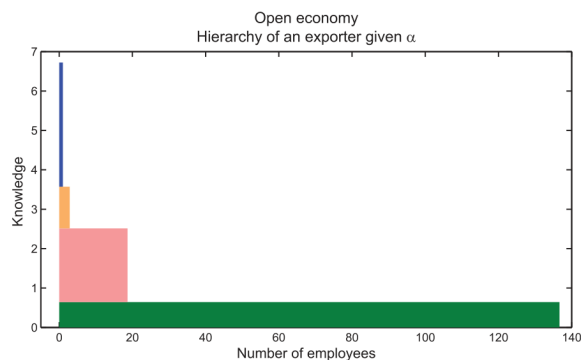
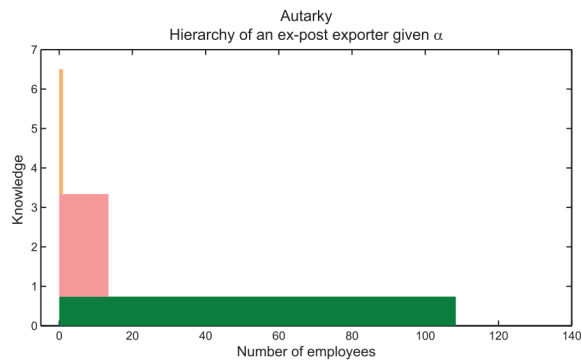


FIGURE VIII
Continued

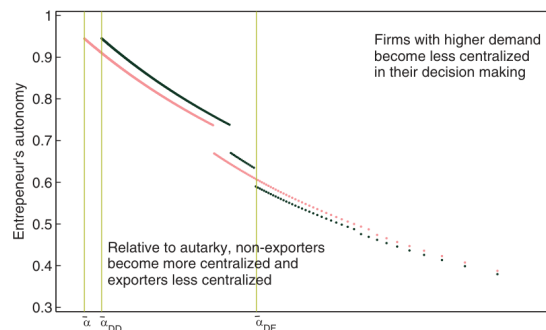


FIGURE IX

Changes in Autonomy before and after a Bilateral Trade Liberalization

Figure VIII, part 2. Exporter hierarchy before and after trade.

Figure IX. Entrepreneur autonomy before and after liberalization.

Read. The exporter adds a management layer and expands lower-layer employment. Figure IX shows entrepreneur autonomy falling with demand and falling for exporters after trade, while nonexporters become more centralized.

Economic message. Large exporting firms decentralize because hierarchy lets managers solve exceptions and frees lower layers from needing broad knowledge.

10.3 Comparative statics and alternative productivity measures

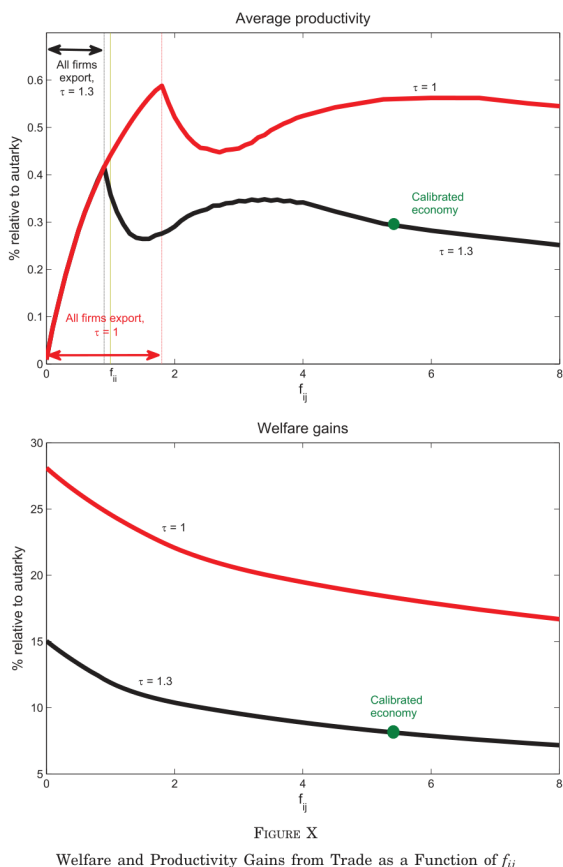


Figure X. Welfare and productivity gains as export fixed cost changes.

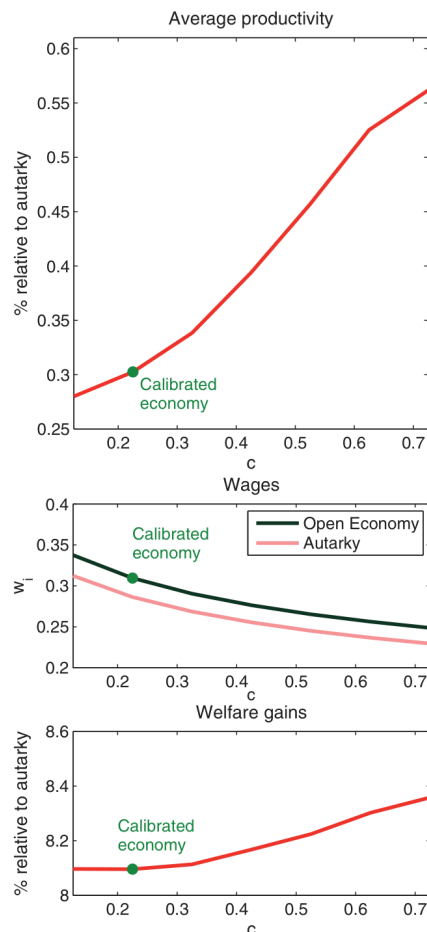


Figure XI. Comparative statics in knowledge cost c .

Read. Productivity gains are zero when all firms export and also disappear as export fixed costs become prohibitive. Welfare gains fall as export fixed costs rise. Higher knowledge costs make organization more valuable and raise organizational productivity gains from trade.

Economic message. The organizational gain from trade is strongest when trade changes firm scale selectively, and when internal organization is important for economizing on knowledge.

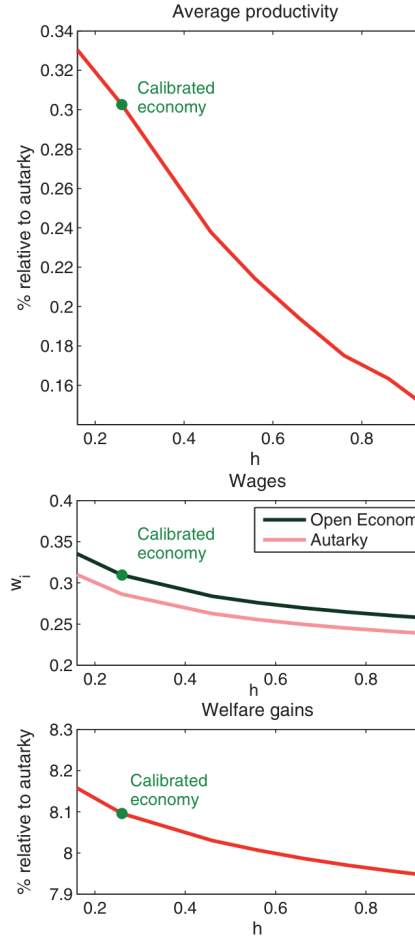


Figure XII. Comparative statics in communication cost h .

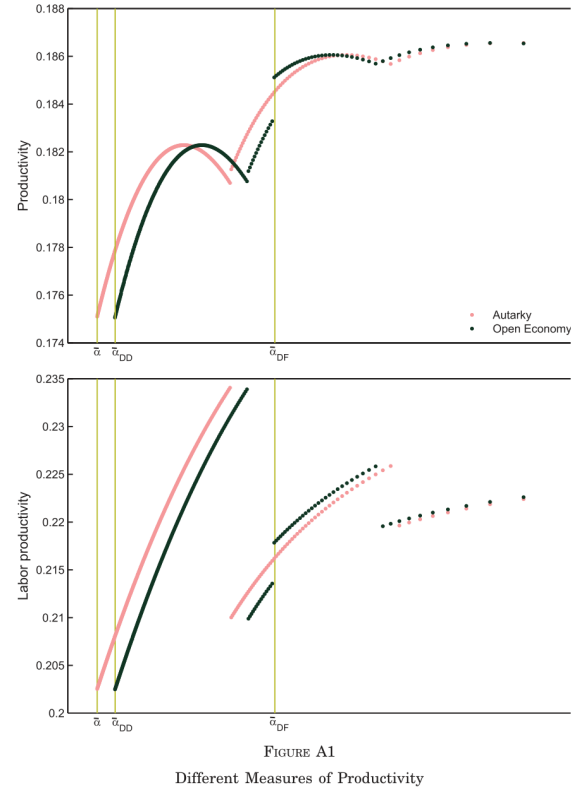


Figure A1, part 1. Physical and labor productivity measures.

Read. Higher communication costs reduce productivity and welfare gains because managerial knowledge is harder to leverage. Appendix Figure A1 shows that productivity patterns depend on the measure used, but exporters remain more productive on average.

Economic message. Communication technology shapes how much trade can improve organization. Empirical productivity measures matter because revenue-based measures confound physical efficiency and price effects.

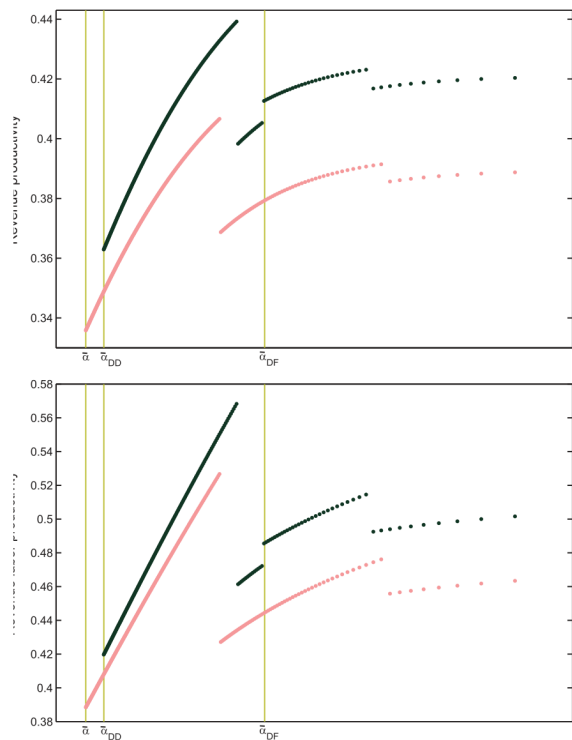


FIGURE A1
Continued

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TABLE A1
PRODUCTIVITY GAINS RELATIVE TO AUTARKY

Weight	Productivity				Revenue productivity			
	1	$n(\alpha)$	$r(\alpha)$	$q(\alpha)$	1	$n(\alpha)$	$r(\alpha)$	$q(\alpha)$
All firms	0.03%	0.30%	0.27%	0.22%	8.16%	8.63%	8.52%	8.47%
Exporters	0.10%	0.04%	0.05%	0.05%	8.33%	8.22%	8.21%	8.22%
Nonexporters	-0.08%	-0.18%	-0.20%	-0.21%	7.95%	7.87%	7.92%	7.89%
Marginal firm		1.00%				1.82%		

Weight	Labor productivity				Revenue labor productivity			
	1	$n(\alpha)$	$r(\alpha)$	$q(\alpha)$	1	$n(\alpha)$	$r(\alpha)$	$q(\alpha)$
All firms	0.08%	0.35%	0.30%	0.28%	8.21%	8.65%	8.54%	8.53%
Exporters	0.33%	0.13%	0.12%	0.13%	8.63%	8.30%	8.29%	8.29%
Nonexporters	-0.03%	0.02%	0.09%	0.08%	8.00%	8.10%	8.25%	8.21%
Marginal firm		2.00%				2.83%		

Figure A1, part 2. Revenue-based productivity measures.

Table A1. Productivity gains relative to autarky.

Read. Revenue productivity and revenue labor productivity rise much more than physical productivity because prices and wages change. Table A1 reports gains by all firms, exporters, nonexporters, and the marginal exporter under alternative weighting schemes.

Economic message. The marginal exporter gains under every productivity metric, but empirical work must distinguish true quantity productivity from revenue productivity.

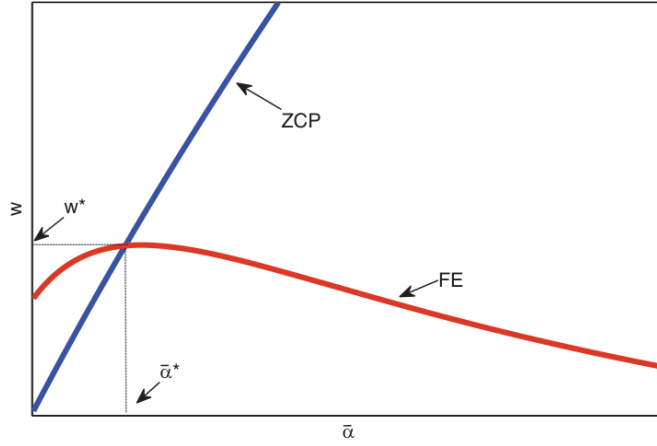


FIGURE B1
Equilibrium w^* and $\bar{\alpha}^*$

Figure B1. Equilibrium wage and cutoff in the proof of existence.

Read. Figure B1 plots the zero-cutoff-profit schedule and the free-entry schedule. Their intersection determines the equilibrium wage and demand cutoff.

Economic message. The appendix establishes that the organizational trade model has a well-defined stationary equilibrium, so the comparative statics and calibration rest on a coherent general equilibrium structure.

11 Short summary

- Firms differ in demand, not in exogenous productivity.
- Demand differences create scale differences.
- Scale differences create organizational differences.
- Organizational differences create productivity differences.
- Trade changes scale: exporters expand, nonexporters contract.
- Exporters tend to add layers and decentralize; nonexporters tend to drop layers and centralize.
- Because organizational choices are discrete, productivity responses are heterogeneous and non-monotone.
- Quantitatively, endogenous organization increases gains from trade by about 41% relative to the comparable standard model.

12 Conclusion

12.1 Core conclusion

Trade affects productivity not only through selection across firms, but also through reorganization within firms. Exporting expands the scale of successful firms, making additional management layers worthwhile. These layers allow firms to economize on knowledge, change the distribution of knowledge and wages inside the firm, and raise productivity for many exporters.

12.2 Incremental contribution revisited

The paper’s key contribution is to make organization endogenous in a trade model. This changes the interpretation of firm productivity: productivity is not only a primitive; it is partly an equilibrium outcome of demand, scale, knowledge costs, and communication costs.

12.3 Economic intuition

A small firm cannot justify a large hierarchy. A large exporter can. When trade opens, exporting firms gain access to a larger market, scale up, and reorganize. The reorganization lowers average cost at the new scale. Nonexporters face more competition, shrink, and may simplify their hierarchy.

12.4 Limitations

- The quantitative exercise is a calibrated structural model rather than direct causal estimation.
- The model abstracts from many real-world dimensions of organization, including multiproduct production, explicit worker heterogeneity, firm-specific management ability, contracting frictions, and short-run adjustment frictions.
- The model’s strong predictions require matched employer-employee data with occupations and wages to be tested directly.
- The symmetric two-country calibration is useful for isolating mechanisms, but real trade liberalizations are asymmetric across countries and sectors.

12.5 Takeaway

The paper turns the firm from a black-box technology into an organization that chooses knowledge, layers, and spans of control. This matters because trade changes not just which firms survive and export, but how firms are internally built. The organizational response is a quantitatively important source of the gains from trade.